Prediction of Chronic Disease Risk in Older Adults using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of older adults. Five waves of socio demographic, clinical, nutritional, psychological, and behavioural data, with selected variables from the externally produced Research and Development Corporation (RAND) + Harmonised TILDA dataset, together were systematically cleaned and standardised using a multi stage preprocessing pipeline. Predictive models, including (*models x, y, z*) were developed to estimate the risk of chronic disease and FR related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, Frailty (FR), chronic disease, nutrition, The Irish Longitudinal Study on Ageing (TILDA)

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

ADL	Activities of Daily Living
AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BP	Blood Pressure from Sphygmomanometer
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
CRP	C-Reactive Protein
CSV	Comma Separated Values
CV	Cross Validation
CVD	Cardiovascular Disease
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
F1	F1-score (harmonic mean of precision and recall)
FI	Frailty Index
FOF	Fear of Falling
FR	Frailty
G2G	Gateway to Global Ageing Data
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
GRIP	Grip Strength

GSR	Gait Speed Reserve
HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength
IADL	Instrumental Activities of Daily Living
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MMSE	Mini-Mental State Examination
MOCA	Montreal Cognitive Assessment
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PC1	First Principal Component
PCA	Principal Component Analysis
PMF	Pseudonymised Microdata File
PRISMA	Preferred Reporting Items for Systematic reviews and Meta Analyses
RAND	Research and Development Corporation
Recall	Recall (true positive rate)
RF	Random Forest
SCQ	Self Completion Questionnaire
SHAP	Shapley Additive Explanations
SLR	Systematic Literature Review
SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
SR	Systematic Review
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)

List of Abbreviations

UGS Usual Gait Speed

WHO World Health Organization

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [7–9]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [10]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [11]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland's *Healthy Ireland* framework, highlight nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [12], Physical Activity (PA) [13], Scales and Questions on Mental Health and Wellbeing (MH) [14], and Social Variables (SOC) [15] improve predictive performance. Estimating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

The Irish Longitudinal Study on Ageing (TILDA)[1] provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced Research and Development Corporation (RAND) Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutrition factors and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

[1] TILDA's dataset are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to University of Limerick (UL) S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [12, 16].

- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as Frailty (FR) progression [6]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$.

  This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:**Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthiness AI in healthcare [17].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).

- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.

- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.

- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and other MH features complement established clinical measures to enhance predictive performance.

- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.

- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis outline

Chapter 2 presents a literature review on ML prediction of chronic disease in ageing populations, focusing on lifestyle, nutrition predictors and methodological considerations. *Chapter* 3 details the research methodology, including the systematic mapping process, the TILDA study design and data collection. In *Chapter* 4, the technical approach is outlined, including the modelling framework, baseline and advanced ML methods, validation and evaluation metrics against established benchmarks. *Chapter 5* presents the experimental results, including model comparisons and feature importance analysis. *Chapter 6* discusses the implications and future directions of the research.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, Hand Grip Strength (HGS), and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.1.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) is a conditional probability $P(Y \mid X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j, guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [18]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [19]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} \mid \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [20].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X.

2.1.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset, and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling for a enabling for a more robust training validation splits in ML models, and explores richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.2 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [21] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [22] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [13]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by Fear of Falling (FOF) [23]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as Center for Epidemiologic Studies Depression Scale (CES-D) or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited, especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.3 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [12] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [24] found vitamin D deficiency predicted Diabetes Mellitus (DM), musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.3.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.3.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [25] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [26, 27]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.4 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [28, 29]. Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables.

A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [30]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes (here). Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.4.1 Models used in machine learning

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, gradient boosting machines GBM, as well as Deep Neural Network (DNN)s, which generally achieve higher predictive accuracy.

Similarly, [31] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an AUC of 0.78, whereas ensemble methods in [28] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.4.2 Performance and validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [30].

2.4.3 Explainability in machine learning for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like Shapley Additive Explanations (SHAP) can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 7, 28]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.5 Gaps in the literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults `(50+)`. This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA studying ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [12], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [25] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability

$P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [31] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [28] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where else ensembles or Brain predicted age difference (brainPAD) approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [12] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [7] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [12], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [18] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through listwise deletion, as in [31], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model's outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [4, 28], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.6 Chapter summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, nutrition, and biological ageing. We also examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the feature engineering and modelling strategies implemented in Chapters 3 and 4, including the use of harmonised variables to ensure reproducibility and generalisability for ML modelling.

Chapter 3

Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping. This included defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined [1]. The integrative literature review [32] helped structure and connect findings across studies, while [33, 34] guided the organisation of ideas. In related methodological work, [2] clarified how Systematic Review (SR)s can identify research gaps, and [35] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed, structured yet flexible methodology suitable for ageing in AI, and health research.

3.1 Search strategy

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as “Irish Longitudinal Study on Ageing", “machine learning,"“chronic disease", “nutrition", “diet", and “physical activity", as shown in Table 3.1.

The search covered studies published from 2011 to 2025. Figure 3.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

Table 3.1: Database searches and number of studies per database 3.3 [1].

Database	**Search**	**Results**
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR “physical activity")	1081
	(TILDA OR “Irish Longitudinal Study on Ageing") AND (“machine learning" OR “predictive model" OR “risk prediction")	71
IEEE Xplore	TILDA OR “Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND (“machine learning" OR “predictive model" OR “risk prediction")	7
ScienceDirect	(TILDA OR “Irish Longitudinal Study on Ageing") AND (“chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity")	257
	(TILDA OR “Irish Longitudinal Study on Ageing") AND (“machine learning" OR “predictive model") AND (“chronic disease" OR diabetes) AND (nutrition OR “physical activity")	20
PubMed	(TILDA OR “Irish Longitudinal Study on Ageing") AND (“chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity")	27
	(TILDA OR “Irish Longitudinal Study on Ageing") AND (“machine learning" OR “predictive model" OR “risk prediction")	9

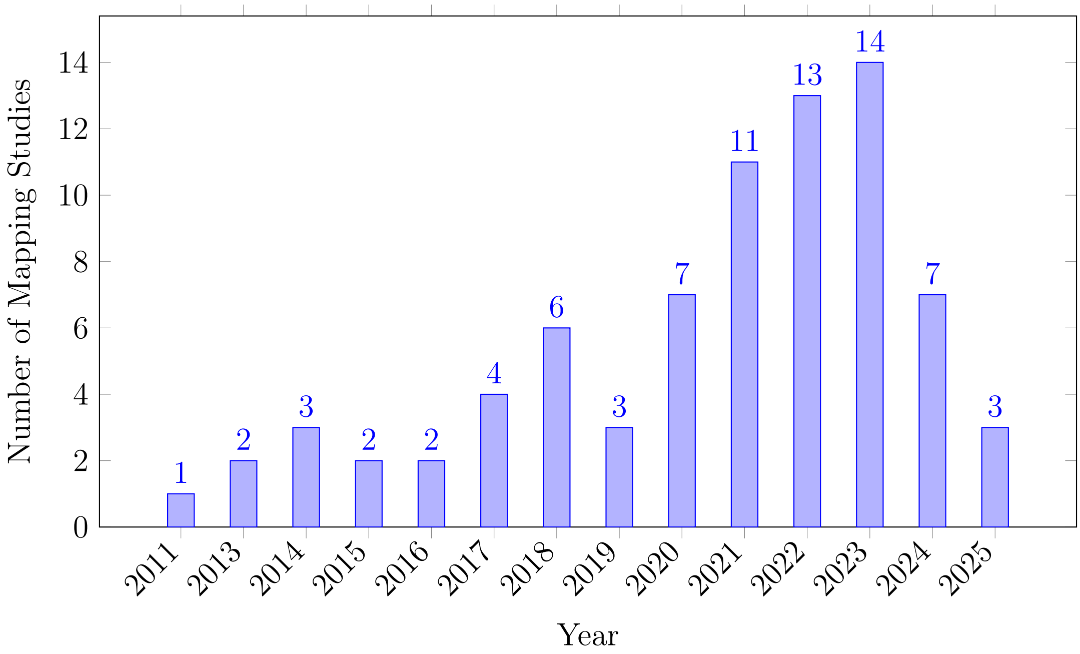

Figure 3.1: Publications per year.

Some search strings returned few or irrelevant entries, particularly for ML applications in TILDA, highlighting a key research gap. Iterations continued until suitable combinations consistently retrieved relevant studies. Duplicates were removed using Zotero's reference management feature. The remaining studies were individually and manually checked for relevance to "ageing," "chronic disease," and ML. Using Zotero's notes functionality, the initially created Table 3.3 captured key elements for each study, including model type, features used, and main findings, as shown in the following Preferred Reporting Items for Systematic reviews and Meta Analyses (PRISMA) flow diagram (Figure 3.2).

3.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customised version of the Newcastle-Ottawa Scale (NOS) was applied (Table 3.2). The NOS evaluates studies across three domains: *selection* (S1–S4), *comparability* (C1–C2), and *outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample,

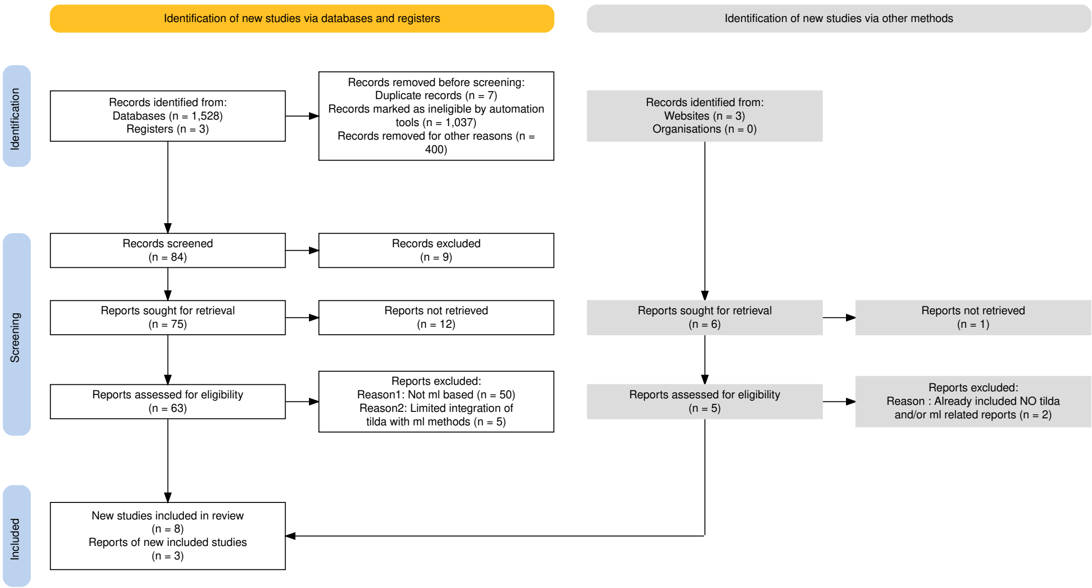

Figure 3.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [5].

clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored $0/1$, recall the Bernoulli random variable ($0 =$ criterion not met, $1 =$ criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: 1. studies critically related to TILDA, 2. studies using TILDA and ML for predictive contexts, which revealed minimal to no integration with TILDA.

Table 3.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

Table 3.2: Quality evaluation of Systematic Literature Review (SLR)s [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [28])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [17])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [6])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [7])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [29])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [36])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [37])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [18])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [25])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [38])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [39])	1	1	1	1	0	0	1	1	1	7

3.2.1 Synthesis and integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by *DoctoralWriting* resources [40, 41], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML applications with TILDA, additional studies on older adult populations and chronic disease were included.

3.3 Study population and design

TILDA provides longitudinal data across five waves (Waves 1–5) for community dwelling adults in Ireland. For benchmarking against international cohorts, an

Table 3.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [18]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [17]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were <= 0.69 and PRAUC <= 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
FOF	Zarudsky (2014) [30]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [20]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), HGS, physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [23]	Wave 1–2	Falls, FOF, HADS-A, CES-D	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [16]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [12]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [25]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability >65 years, self reported
ML	Xu (2023) [28]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, DNN; SHAP explainability	AUC=0.854 for 2 year DM prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [7]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [38]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [29]	Wave 3	brainPAD; processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

external harmonised TILDA dataset was also inspected and prepared. The harmonised dataset, which has been pre aligned to international standards e.g., the Gateway to Global Ageing Data (G2G), contains a subset of variables mapped to a common cross study format and thus differs from the raw TILDA files. Although the harmonised dataset was simpler to process, it was still subjected to the variable filtering and numerical standardisation steps developed as part of this research to ensure compatibility with the five wave cleaned datasets prior to modelling.

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering occurred primarily at the *variable level* during the manual screening stage, which involved features removed for extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.

- **Usable data requirement:** Participants were excluded only if all retained variables were missing for that individual after manual variable screening.

- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.

- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications, differing from exclusion heavy cohort definitions often used in traditional epidemiological studies [42].

3.4 TILDA data collection and screening

TILDA data collection occurred over five longitudinal waves (Table 3.4).

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.

2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.

3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centres or at home (Wave 1 and Wave 3).

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Table 3.4: Dimensions of raw TILDA and harmonised datasets (prior to cleaning Section 4.4).

Dataset	**Collection Period**	**Participants (N)**	**Raw Variables (N)**
Wave 1	2009–2011	8501	1619
Wave 2	2012–2013	7206	724
Wave 3	2014–2015	6397	1029
Wave 4	2016	5713	990
Wave 5	2018–2019	4978	983
Harmonised	2016 (release)	8501	572

3.5 Thematic domains

Variables were organised into seven thematic domains based on ageing literature and study objectives. Each variable's prefix, as defined in the Pseudonymised Microdata File (PMF) metadata (Table 3.5), indicates its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and machine learning.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code (e.g., `ph121`). Variables with multiple responses or loops include an underscore and number to specify the response or iteration (e.g., `ph201_05`).

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area (Table 3.5). These prefixes provide a consistent framework for categorising features and guiding the assignment into thematic domains for downstream analysis.

By categorising the raw variables into these seven thematic domains, a structured foundation was established for the technical dimensionality reduction and cleaning pipeline described in the following Chapter 4.

Table 3.5: PMF Derived variable prefixes grouped by thematic domains [3].

Variable Prefix	Research Area
1: Socio Demographic & Context	
INC	Income variables
SES	Social class
SOC	Social
2: Physical Health & Clinical Baseline	
BP	Blood pressure from sphygmomanometer
MD	Medication data
SCR	Screening tests
3: Functional Status & Frailty	
ADL	Activities of daily living
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
GRIP	Grip strength
IADL	Instrumental activities of daily living
4: Lifestyle & Behavioural	
BEH	Health behaviours
5: Nutrition & Metabolic Biomarkers	
BLOODS	Results of blood sample analysis
6: Psychological & Cognitive	
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
MH	Scales and questions on mental health and wellbeing
Other / Administrative	
G2G	Gateway to Global Aging Data

Chapter 4

Technical Implementation

Prior to any data cleaning or modelling, a structured multi-stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. As shown in Table 4.1 (for example, 1619 $\rightarrow$ 809 variables in Wave 1), the raw PMF files were reduced to a subset of variables of interest. This procedure also separated automated diagnostic profiling from manual variable exclusion, ensuring transparency in all selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (`.sav`) files, while the harmonised dataset was loaded from a (`Stata`) (`.dta`) file due to technical compatibility issues.

2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds detailed in the next Section 4.1. Variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.

4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained `variables of interest`.

5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No cleaning, recoding, or imputation was applied at this stage.

4.1 Manual variable filtering workflow

Python scripts were used to profile all variables across each wave. For every variable, the following metrics were computed: data type, number of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format to support structured manual review prior to variable dropping.

Manual filtering was applied according to the following criteria:

- **Missingness:** Drop variables with $> 80\%$ missing values. This threshold balances retaining sufficient feature coverage while ensuring statistical reliability. Variables exceeding this threshold provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.

- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.

- **Dominant category (top frequency):** Variables where a single category (sample value) occurs in over $> 90\%$ of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.

- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave's PMF file into Comma Separated Values (CSV) files (Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

During manual screening and filtering, variables exceeding exclusion thresholds (e.g., high missingness, single value, dominant category, administrative identifiers) were visually flagged (e.g., marked in red). Non flagged variables were programmatically concatenated using spreadsheet formulas (e.g., `=TEXTJOIN(", ", TRUE, J2:Jn)`) and then manually transferred into wave specific JSON files for each wave. These files represent the *variables of interest* retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing and cleaning in Section 4.4.

The retained variables, defined in JSON files, were subsequently used to subset the raw datasets. As shown in Table 4.1, this process yielded a reduced dataset for each wave containing only the variables of interest. At this stage, no cleaning or recoding was performed. The operation served purely as dimensionality reduction, preparing the data for the automated six step scenario cleaning pipeline described in Section 4.4.

Importantly, all variable exclusion decisions were completed at this stage. Subsequent preprocessing steps focus exclusively on numerical standardisation, recoding, and preparation for machine learning rather than further feature removal. With the variables of interest selected and the datasets reduced to a manageable dimensionality, the next stage focuses on automated preprocessing, recoding, and preparation for ML modelling.

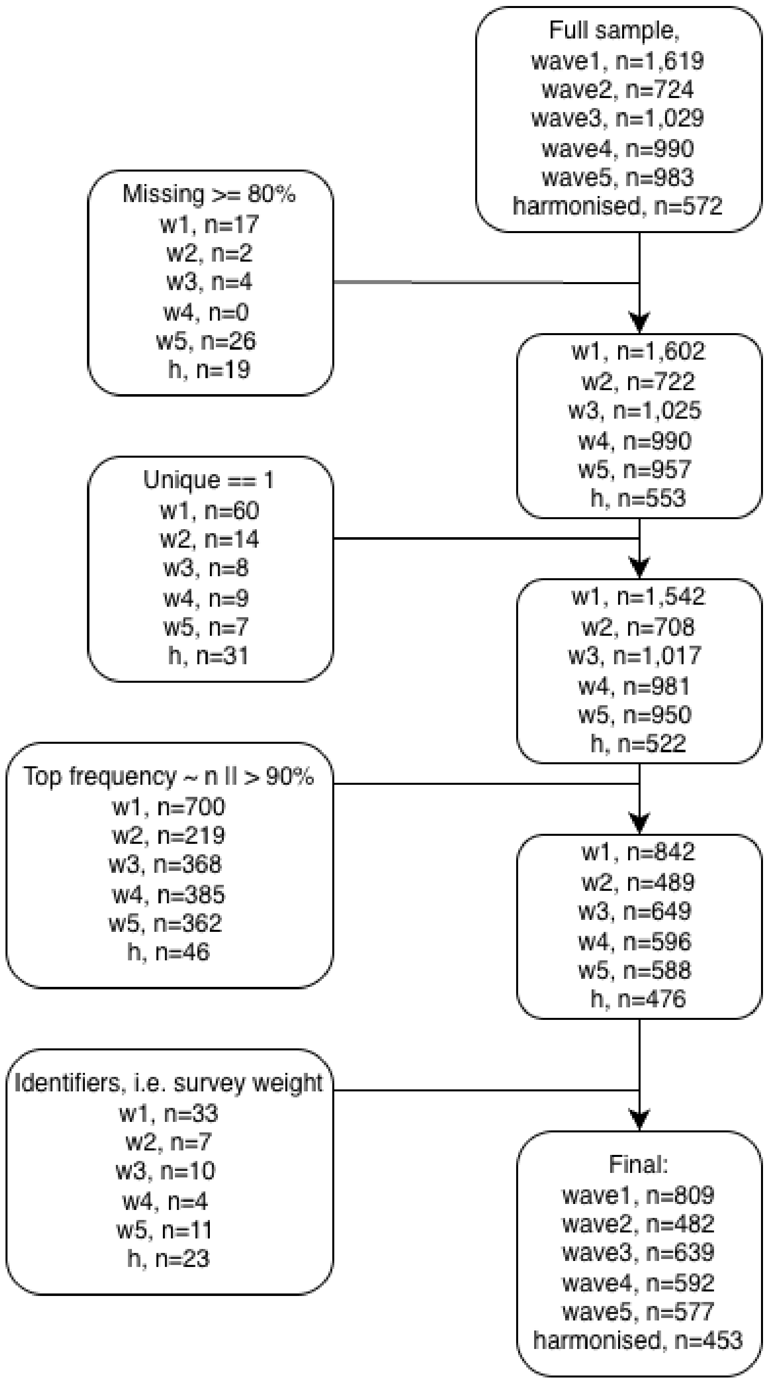

Figure 4.1: Flowchart of the five step workflow for variable screening and filtering. Placeholder (n) counts indicate the progress of remaining variables per dataset from raw to filtered datasets after each step, ready for automated preprocessing.

4.2 Reduced dataset overview

To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`.ipynb`) file implemented with `pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves. The purpose of this profiling stage was not transformation but to expose structural inconsistencies and latent formatting issues that would undermine direct numerical conversion.

Table 4.1: Input shape for cleaning pipeline (post manual filtering workflow 4.1).

Dataset	Row (N participants)	Column (N features)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised	8501	453

For example, in wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns. Age distributions confirmed that the study population was restricted to older adults across waves, however, ages were bottom coded for participants younger than 50 years and top coded for those aged 65–85 years and above. Missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

Despite this, the exploratory visualiser was limited because the raw data were not yet cleaned, many outputs were incorrect or nonsensical due to hybrid formats and inconsistent coding. For instance, inspection revealed that many variables across both the wave specific and harmonised datasets, while appearing numeric at first glance, in fact contained hybrid representations combining numeric prefixes with descriptive text (e.g., “1.Rarely or none of the time (less than 1 day)”, "4.All of the time (5–7 days)", or prefixed labels such as "0.not coupled", "1.coupled"). A trivial string stripping approach would therefore produce incorrect numeric values (for example, reducing "5–7 days" to `5`) resulting in silently introducing semantic distortion. Similarly, numeric columns sometimes contained sentinel codes (e.g., -98, -99) or embedded symbols ('$', '*cm*', '%') that skewed visual summaries.

These insights, revealed through systematic profiling, guided the development of the subsequent cleaning pipeline steps, ensuring consistent numeric logic representations (e.g., midpoints for ranges, standardised sentinel handling).

4.3 Thematic grouping of retained variables

Using the PMF prefixes described in Table 3.5 as a guide, variables were assigned to seven thematic groups (Table 4.2), each capturing a conceptually coherent domain: physical, cognitive, social, and behavioural aspects of ageing. This structure supports reproducible preprocessing and helps machine learning models focus on meaningful, longitudinally robust features.

Specifically, the thematic grouping:

1. Provides a clear framework for automated cleaning and filtering rules.
2. Ensures that model input variables are conceptually aligned with study objectives.

The seven thematic groups integrate information from CAPI, SCQ, and derived variables, covering sociodemographic context, physical health, functional status, lifestyle behaviours, nutrition and metabolic biomarkers, psychological and cognitive measures, and administrative identifiers.

The gradient *heatmap* integrated at the bottom of Table 4.2 visually illustrates the relative dominance of each thematic domain across the five waves. Darker blue shading indicates domains with a higher number of variables, providing a quick summary of domain coverage and prominence. This integrated visualisation supports interpretation by highlighting which domains contribute the most features to the dataset.

Table 4.2: Thematic domain grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variable Examples	Total Features
1	**Socio-Demographic & Context**	Baseline demographic and socioeconomic indicators	Age/Sex Education Income Marital status	844
2	**Physical Health & Clinical**	Clinical and self reported health measures	Blood pressure BMI Chronic disease status Self rated health	839
3	**Functional Status & Frailty**	Physical functioning and frailty markers	Activities of Daily Living (ADL)/Instrumental Activities of Daily Living (IADL) Grip strength Gait speed	174
4	**Lifestyle & Behavioural**	Health behaviours and activity patterns	Physical activity Smoking Alcohol use	120
5	**Nutrition & Metabolic**	Metabolic and biomarker indicators	C-Reactive Protein (CRP) Creatinine Lipid markers	22
6	**Psychological & Cognitive**	Cognitive performance and mental health	Mini-Mental State Examination (MMSE) Depressive symptoms Stress scale	291
7	**Other / Administrative**	Survey metadata and structural variables	Sampling weight Completion status Contact names	809

	SocioDemographic_Context	PhysicalHealth_Clinical	FunctionalStatus_Frailty	Lifestyle_Behavioural	Nutrition_Metabolic	Psychological_Cognitive	Other_Administrative	Total_Features
wave1	272	149	48	58	10	97	175	809
wave2	119	148	18	21	0	38	138	482
wave3	164	184	44	14	8	93	132	639
wave4	161	186	30	13	4	36	162	592
wave5	128	172	34	14	0	27	202	577
Total_per_Domain	844	839	174	120	22	291	809	3099

Thematic domain dominance across waves (higher n features per domain).

4.4 Automated cleaning engine

The core challenge in converting heterogeneous variable types across the six datasets (Table 4.1) was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained features using a single numerical representation (`float64`) with a unified missing or non response sentinel (`-1.0`). The exploratory profiling across all waves showed that categorical variables were dominant overall ($2,916$ columns), while numeric variables were fragmented across multiple types, with `float64` being the most prevalent native numeric representation (462 columns), motivating its use as a common standard to ensure consistency and compatibility with downstream ML models, while accommodating decimal values arising from range midpoints, frequency normalisation, and ordinal scaling that cannot be reliably represented using integer types. To address this, the solution was implemented as a two stage system. The first stage performs automated *variable scenario detection*, identifying the structural form of each variable based on observed value patterns using the `detect_column_scenario_mapping` function (Appendix C). This stage includes additional diagnostic categories used during development to surface unhandled edge cases.

The second stage applies a deterministic *six step numerical cleaning engine* (Appendix E). While nine scenario categories were initially defined during development, three served exclusively as validation fallbacks and were progressively eliminated as detection coverage improved, with a full scenario to step mapping counts provided in Appendix B. The final pipeline therefore consists of six active transformation steps, applied sequentially and uniformly across all waves.

4.4.1 Stage 1: Variable scenario mapping and validation

Stage 1 focused on automatically identifying how each variable should be handled during cleaning. The automated detection function in Appendix C was developed to classify variables based on their observed value structure, including the presence of sentinel codes, numeric annotations, ordinal labels, and mixed string numeric formats. While multiple intermediate detection categories were used during development, these served solely as internal routing logic. While all variables passed sequentially through the six step cleaning pipeline, most were fully resolved in the earlier steps (Steps 1–5). The final step (Step 6 in Appendix D) was applied only to residual variables that could not be deterministically resolved by earlier transformations.

The additional detection categories (e.g., "single value categorical", "special symbol mixed", and "unknown/complex") were intentionally included as diagnostic fallbacks to surface unhandled edge cases due to the complexity and size of the data. As rules matured, variables initially defaulted to these fallback categories were progressively resolved into earlier, well defined scenarios, leaving none remaining for the final pipeline execution.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of feature states before and after cleaning, including all representative sample values for each variable. This allowed direct inspection of how the same variable was represented across waves prior to cleaning, particularly in cases where identical semantic categories appeared in different orders or formats. By comparing these grouped samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the `show_grouped_vars_for_scenarios` function (Appendix F) was formatted to create an automatic friendly ".md" file for rapid, surface level verification, which honed the final mapping dictionary (an example can be found in Appendix G) for all features across all waves. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.4.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed sequentially by the *six step cleaning engine*, resolving simple and unambiguous cases first to maximise the proportion of clean numerical features before tackling more complex structures, particularly censored and multi class scenarios (Figure 4.2). With later steps progressively shifting from cleaning into semantic feature engineering, this deterministic sequence applied identical transformation and feature encoding logic to each wave, with later steps (particularly Steps 4–6) performing rule based feature engineering by deriving semantically meaningful numerical representations directly from observed category tokens at runtime. The methodological rationale and implications of this integrated engineering which reframes cleaning as semantic feature construction are detailed in Section 4.4.3. (See Appendix D for implementation).

This design reflects the empirical diversity of survey encodings, where a small subset of variables only reveal their semantic structure after simpler transformations have been applied.

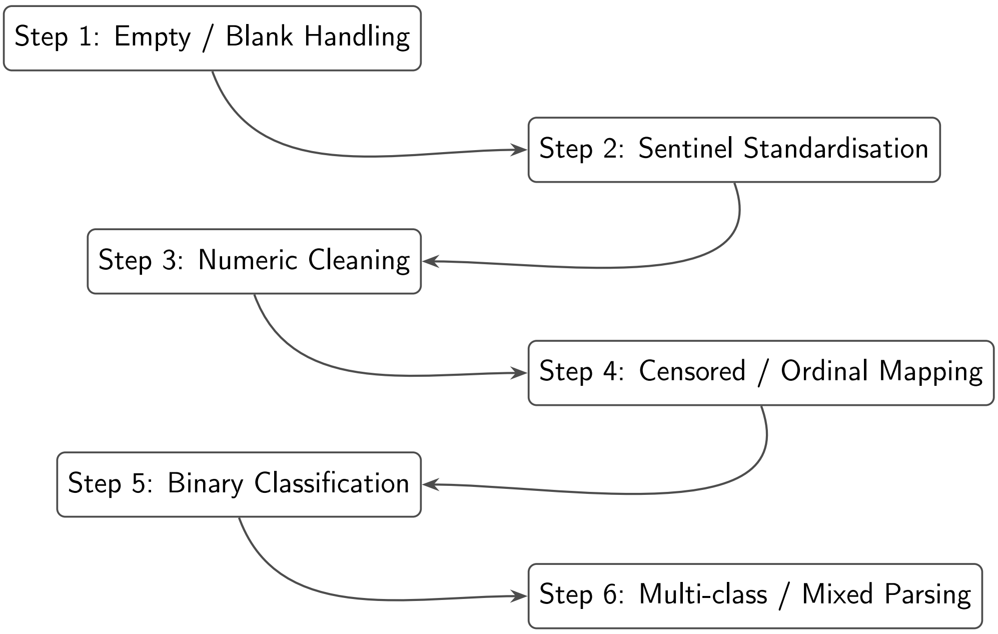

Figure 4.2: Flowchart of the final six step cleaning pipeline applied to all TILDA waves. See full transformation tables in Appendix B.7 and E for the detailed progress "`.ipynb`" script, at each scenario cleaning step.

The final cleaning pipeline consisted of six deterministic steps applied uniformly across all TILDA waves, with increasingly complex transformations handled in later stages. Steps 4–6 required particular attention due to hybrid, ordinal, or multi class formats. A summary of each step and examples is provided below.

1. **Empty / Blank Values:** All missing entries, empty strings, and whitespace were converted to the standard non-response sentinel (`-1.0`).
2. **Sentinel Values:** Wave specific non-response codes (e.g., `-98`, `-99`) and textual sentinels (e.g., `"dk"`, `"not applicable"`) were mapped to the unified (`-1.0`) placeholder.
3. **Numeric Transformation:** Numeric like columns containing embedded units, symbols, or hybrid strings (e.g., `50+`, `<10`, `1-2.5`) were parsed using regex and converted to `float64`. For ranges, midpoints &/or boundary values were used. For censored or annotated values, appropriate upper/lower bounds were applied.
4. **Censored / Ordinal Variables:** This step addressed variables representing ordinal scales, bracketed numeric ranges, or frequency expressions (e.g., "2–3 days a week", "daily"). Transformations included:
   - Converting textual ordinal labels and Likert scale responses to numeric equivalents.
   - Mapping calendar based frequency descriptors to numeric weekly equivalents.
   - Manual verification for low count or unusual variables, supported by the Markdown audit outputs (Appendix G), to ensure semantic meaning was preserved.
5. **Binary Categorical Variables:** Two level features (e.g., Yes/No, Male/Female) were dynamically encoded using mappings derived from the observed value set of each variable:
   - Semantic rules were applied e.g., ("no", "not", "female" $\rightarrow$ 0.0) and ("yes", "male" $\rightarrow$ 1.0).
   - Count based detection verified that variables had exactly two non-missing unique values.
   - Features with more than two sample values were also analysed for sentinel presence:

    - If a variable had three observed values and one was a sentinel, it was treated as binary.
    - If a variable had four observed values and two were sentinels, it was treated as binary.
  - Sentinels were preserved as `-1.0`.
  - This approach ensured order invariant encoding across waves, even when original labels or numeric prefixes differed.
  - *Example (sentinel case):* Feature `fl002_5` with pre transformed values `['NOT Getting in or out of bed', -1.0, 'Getting in or out of bed']` was encoded as `[0.0, -1.0, 1.0]`, demonstrating how a sentinel preserves binary semantics.

6. **Multi-class / Mixed Variables (rule based feature engineering):** Variables with remaining multi class structure, mixed types, or unresolved string numeric hybrids after earlier transformations were processed in the final step.

   Technical implementation details for this step are provided in Appendix D:

   - Lexical parsing extracted numbers, ranges, units, percentages, and textual frequency descriptors into floats, yielding interpretable numerical feature representations.
   - Sentinel handling ensured all known non-response codes mapped to (`-1.0`).
   - Original values were grouped and assigned numeric identifiers in first seen order, non numeric categorical labels were assigned integer codes above the highest observed numeric value to prevent collisions.
   - Manual inspection using the Markdown audit file verified that semantic equivalence was maintained across waves.
   - Step 6 was designed to be *order invariant*, dynamically interpreting category values within each wave rather than relying on predefined lookup tables shared across waves. This preserved consistent numeric

representation regardless of variable ordering or wave specific encodings.

- *Example:* Step 5 handled simple binary variables such as `CHRany_pain` (any pain reported) with pre transformed values `['pain', 'No pain']` and numeric mapping `[1.0, 0.0]`. In contrast, Step 6 dynamically processed multi class variables like `we114` (pre tax wage/salary) many of which contained over 20 distinct pre transformed categories (e.g., `-1.0, '200-299', 'Less than 100', ..., '6,000-6,999'`) into consistent numeric representations (e.g., `-1.0, 249.5, 99, ..., 6499.5`). This illustrates why Step 6 required wave specific, dynamic parsing to maintain semantic equivalence across diverse observed values.

4.4.3 Feature engineering within the cleaning pipeline

Although the six step pipeline is framed primarily as a numerical cleaning procedure, Steps 4–6 also perform explicit rule based feature engineering by interpreting heterogeneous categorical survey responses and converting them into semantically meaningful numerical representations. Rather than acting as a separate post processing stage, feature engineering is embedded directly within the cleaning logic, formalising ordinal structure, frequency, and magnitude encoded in text into directly modelable `float64` values suitable for downstream analysis.

Crucially, this process engineers the *representation* of each feature rather than expanding the feature space. For example, free text frequency responses such as "`once or twice a week`", "`a couple of times per month`", or "`almost every day`" are not treated as nominal categories. Instead, lexical cues (e.g., *once*, *twice*, *couple*) are resolved to numeric quantities and normalised onto a common temporal scale, yielding a continuous rate representation. Implicit semantic information such as ordering, intensity, and frequency is therefore preserved through deterministic parsing and normalisation without introducing additional variables. This design maintains the original dimensionality of the dataset while ensuring numerical homogeneity and semantic integrity, producing a compact and model ready representation of complex real world survey data.

4.4.4 Cross wave semantic verification

Given the absence of guaranteed category ordering across waves, preserving semantic equivalence during numerical encoding and feature representation was essential to avoid introducing distortions into downstream ML modelling. Pre and post transformation values for each group were exported to a readable Markdown (`.md`) file, as shown in Appendix G. This representation confirmed that features of known importance for downstream predictive models were preserved and correctly encoded. For example, features such as `ph001` (self rated general health), `ph009` (comparative self rated health), and `bh006` (smoking frequency) correspond to items identified by [6] as among the most influential predictors in mortality models by the Frailty Index (FI). Retaining these features from the manual filtering thresholds applied in (Section 4.1) and encoding them consistently across waves ensured that the cleaned dataset maintains coverage and semantic integrity, preventing distortions that could potentially compromise model performance.

A key challenge in cleaning the five TILDA waves was the lack of guaranteed ordering for categorical values of the same variable across different waves. Table 4.3 illustrates this for waves 4 and 5.

For example, gender (`sex`) variables appeared under different value encodings, with varied category orderings and numeric prefixes across waves. Values such as `['Male', 'Female']`, `['Female', 'Male']`, and prefixed forms like `[''1.male', '2.female']` were all mapped to a consistent numeric representation (`1.0` for male, `0.0` for female). This ensured that semantic equivalence, rather than positional ordering or wave specific encoding, determined numerical assignment.

The variable `ph001` found in Appendix G corresponds to the survey question: "Now I would like to ask you some questions about your health. Would you say your health is. . . "?. Responses are ordinal (Poor to Excellent) with non-response categories.

The grouped Markdown representation reveals three critical points:

1. Semantically identical categories are correctly grouped even under different

variable names (`ph001` vs `ph009`), enabling direct comparison of representations that would otherwise appear fragmented across waves.

2. Numerical encoding is invariant to category ordering, preserving ordinal meaning across waves. Without this approach, positional encoding would assign inconsistent values, potentially distorting performance for downstream models.

3. For variables with a large and heterogeneous set of observed values (e.g., income bands, frequency ranges, or mixed numeric text responses), the grouped view provided essential visibility into how complex category spaces were being collapsed into stable numeric representations, supporting both validation and error detection at scale.

Table 4.3: Inconsistent category ordering for the same feature *ph*009 (`self rated health compared to other people your age`) across waves and its impact on positional (order based) label encoding.

Pos	Wave 4		Wave 5	
	Category	**Code**	**Category**	**Code**
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

4.5 Validation of the cleaning pipeline

The purpose of the cleaning validation stage was to assess whether the automated numerical parsing and encoding pipeline preserved meaningful biological and statistical structure within the cleaned datasets. Rather than asserting correctness based solely on successful execution, validation was performed by evaluating whether known relationships reported in the literature could be recovered using the cleaned data under realistic data governance constraints.

Due to restricted access to certain outcome variables within TILDA, full end-to-end replication of all published models was not always feasible. In particular, access through the `Hotdesk Facility` for the official mortality outcomes is subject to additional governance requirements, including institutional GDPR training certification and approval by the TILDA data access committee [43].

As a result, validation focused on two complementary strategies:

(i) Internal consistency checks against established biological and clinical relationships, and

(ii) Constrained replication of published modelling frameworks using available predictors and the derived `8y_mortality_proxy` outcome.

Together, these validation steps demonstrate that the cleaning pipeline retains expected relational patterns and produces datasets appropriate for downstream modelling tasks.

4.5.1 Internal feature relationships

The first validation stage assessed whether cleaned predictors preserved biologically and clinically plausible relationships reported in prior TILDA studies. This analysis focused on predictors identified by [4] as accounting for approximately 95% of the explained variance in Gait Speed Reserve (GSR), defined as the difference between maximum and usual gait speed.

As GSR is not directly available in the public wave 3 dataset, validation was performed by examining the directional relationships between the 7 primary predictors themselves. This approach evaluates whether the numerical encoding process, including sentinel handling, missing value treatment, and ordinal mapping, retained the expected internal structure of the data rather than reproducing outcome specific performance.

Figure 4.3 presents the correlation matrix for the selected predictors following cleaning. Grip strength exhibited a negative association with both age and chair stand time, reflecting the inverse relationship between muscular strength and physical performance decline. Cognitive performance (Montreal Cognitive Assessment (MOCA)) demonstrated a moderate negative correlation with age and a positive association with educational level, consistent with known patterns of cognitive reserve in older adults. Education correlated positively with cognitive performance, and BMI showed the expected weak negative association with age, reflecting reduced physiological reserve in older cohorts.

Table 4.4 summarises the expected and observed directional relationships between key predictor pairs. Notably, grip strength showed moderate negative correlations with both age ($r = -0.265$) and chair stand time ($r = -0.170$), reflecting the expected inverse relationship between muscular strength and physical performance decline. Cognitive performance (MOCA) displayed a moderate negative association with age ($r = -0.381$) and a positive association with education ($r = 0.367$), consistent with known patterns of cognitive reserve in older adults. BMI demonstrated the anticipated weak negative correlation with age ($r = -0.073$). These results confirm that the cleaning pipeline retained the expected numerical structure and relational patterns of the predictors, rather than introducing artificial associations.

4.5.2 Model based validation

To evaluate whether the cleaned dataset preserves predictive utility, a constrained replication of [6] was conducted. The resulting predictive performance for both model variants and sexes is shown in the following Figure 4.4. The study predicted 8-year mortality using 32 FI items. In the cleaned wave 1 dataset following the deterministic six step cleaning pipeline described in Section 4.4.2, only 16 of these 32 FI items remained following the manual variable filtering described in Section 4.1, which prioritised global numeric consistency across waves.

An '`8y_mortality_proxy`' outcome was constructed to approximate the sex specific 8-year mortality rates reported in [6] (men: 14.9%, women: 11.1%), derived from linkage to Ireland's Central Register Office records [44]. Due to access

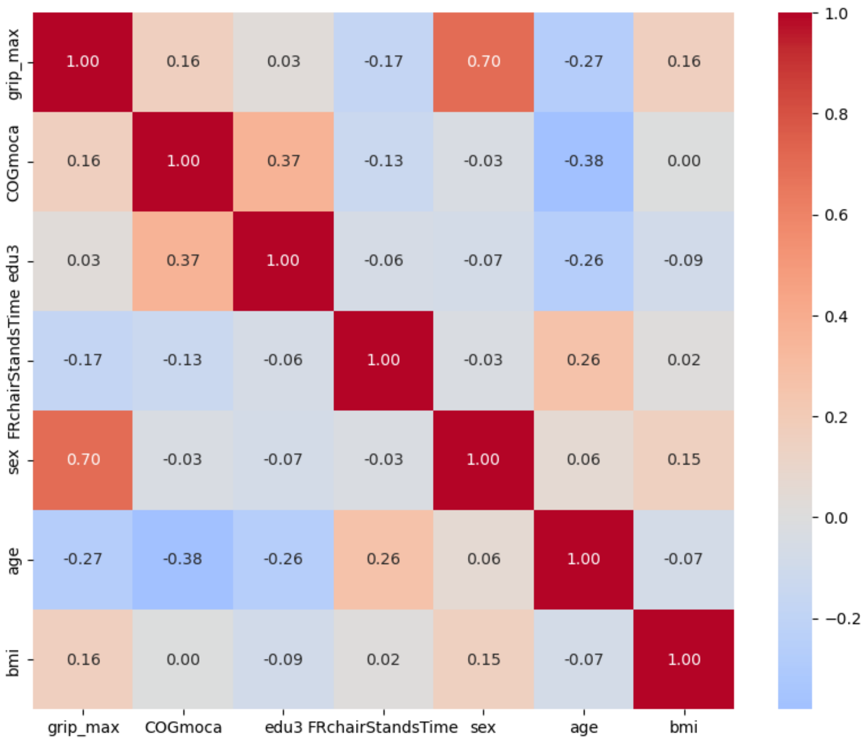

Figure 4.3: Correlation heatmap for top 7 predictors (Wave 3).

Table 4.4: Directional validation of the 7 features explaining 95% of GSR variance as identified by [4].

Predictor 1	**Predictor 2**	**Expected Direction / Justification**	**Observed** r
Grip Strength (GRIP)	Age	**Negative (-)**: GRIP is a positive predictor of GSR, while age is a negative predictor.	-0.265
GRIP	Chair Stands Time	**Negative (-)**: Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests.	-0.170
MOCA Score	Age	**Negative (-)**: Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve.	-0.381
Education	MOCA Score	**Positive (+)**: Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR.	0.367
BMI	Age	**Weak / Mixed**: BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak.	-0.073

restrictions, the official mortality data were unavailable. Therefore, mortality status was first assigned by randomly selecting deceased participants from the wave 1 cohort aged 50 years or older to match the published sex specific mortality proportions observed at wave 5 follow up. An equal number of surviving participants were then randomly sampled within each sex to construct balanced datasets for model training. The resulting random proxy datasets included 1,114 men (557 deceased) and 982 women (491 deceased), reflecting the baseline wave 1 population aged ≥ 50 years ($n = 8,172$: 3,743 men and 4,429 women).

Two variants of this proxy were implemented:

(i) the purely random assignment described above, used as a structural baseline, and

(ii) biologically informed probabilistic assignment that leverages latent frailty captured via Principal Component Analysis (PCA). This dual construction allowed separation of statistical artefact from biologically meaningful signal.

The random proxy provides an initial diagnostic benchmark, mortality labels were assigned using purely random sampling within each sex to match the published 8-year mortality proportions. While this preserves overall class balance, it does not encode any biological or frailty related information and therefore serves only as a structural validation baseline.

For the biologically informed proxy, the 16 available FI items were standardised to zero mean and unit variance using z-score transformation. PCA was then performed separately within each sex on complete case observations to extract the principal component representing the dominant latent frailty structure. The First Principal Component (PC1) was retained as a latent frailty score, and its direction was aligned such that higher PC1 scores correspond to higher frailty by enforcing a positive correlation with the calculated FI. This latent frailty score was subsequently combined with age to construct a probabilistic mortality assignment reflecting both biological vulnerability and time dependent risk.

Two model variants were tested for each sex:

1. **FI only model:** using the 16 available FI items, without age.

2. **FI + Age model:** including age as an additional predictor.

Model training and evaluation followed the procedure reported in [6] as closely as permitted by the available data. Features were min–max normalised prior to modelling. For each sex, data was split into training (80%) and testing (20%) sets. To mitigate overfitting and obtain robust estimates of model performance, 5-fold stratified cross-validation was applied on the training data. AUC was selected as the primary evaluation metric due to its widespread use in prior TILDA and ML based studies (Figure 3.2) and its robustness to class imbalance.

For six FI items (`mh0014`, `ph001`, `ph102`, `ph108`, `ph114`, `bh201`), the cleaned dataset transformations differed from the scoring rules published in [6]. To align precisely with the original 16/32-item FI methodology, these items were recoded following the published ordinal and dichotomous mappings. This step was necessary to ensure that the six remapped FI items reflects the same interpretation of health deficits used in the literature and is consistent with standard FI construction principles [45]. Models were then re-evaluated using this partial remapping FI to assess sensitivity to feature encoding.

The resulting AUC values are reported separately for (i) cleaned (Figure 4.4) vs 6/16 recoded FI items (Figure 4.5), (ii) random vs PCA informed mortality assignment, (iii) FI only vs FI + Age models, and (iv) men and women.

These results highlight several key points:

1. **Random proxy baseline (cleaned vs recoded FI):** Under purely random mortality assignment, discrimination remained at chance level for both versions of the FI. For the cleaned 16 FI items, the FI only model achieved AUC = 0.507 (men) and 0.512 (women), increasing marginally to 0.516 (men) and 0.531 (women) when age was included. When six of the sixteen available deficits were recoded to align with the published FI scoring rules [6], the AUC values increased to 0.567 (men) and 0.537 (women) for the FI only model, and to 0.572 (men) and 0.605 (women) for the FI + Age model. Despite this increase relative to the fully cleaned encoding, discrimination remained close to chance under random label assignment.

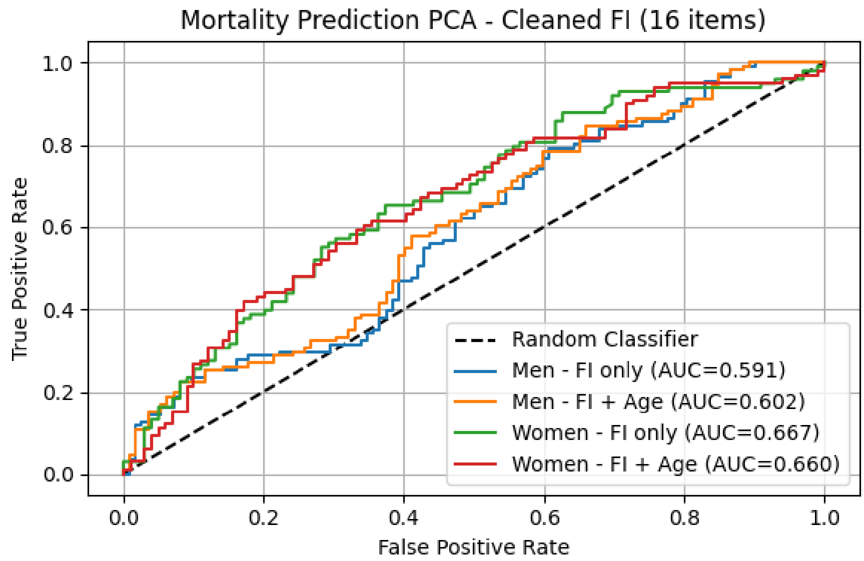

Figure 4.4: ROC curves for mortality prediction using the cleaned 16 FI items under the PCA informed mortality proxy. Separate models are shown for men and women, with and without age.

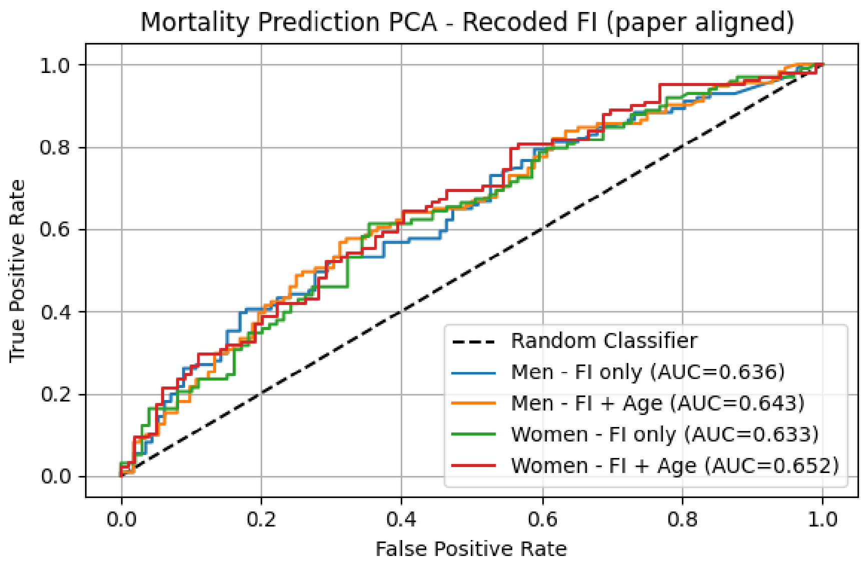

Figure 4.5: ROC curves for mortality prediction using the 6/16 recoded FI items aligned with [6]. Performance is shown separately for men and women, with and without age.

These results confirm that, without embedding biological frailty structure into the outcome, predictive performance reflects statistical noise rather than meaningful signal, regardless of encoding refinement.

2. **PCA informed mortality proxy (cleaned vs recoded FI):** When mortality assignment incorporated the PCA derived frailty representation PC1, discrimination improved relative to the random baseline for both FI constructions. For the 16 cleaned FI items, the FI only model achieved AUC = 0.591 (men) and 0.667 (women), increasing to 0.602 (men) and 0.660 (women) when age was included. When six of the sixteen deficits were recoded in accordance with the published scoring framework, the FI-only model achieved AUC = 0.636 (men) and 0.633 (women), while the FI + age model achieved 0.643 (men) and 0.652 (women).

   This indicates that closer alignment with published ordinal deficit scoring improves discriminative stable performance relative to the simplified numeric encoding applied during initial cleaning.

3. **Comparison with published performance:** Absolute AUC values remain below those reported in [6] (approximately 0.70 for FI only and up to 0.80–0.90 when age is included), which is expected given the use of a constructed mortality proxy and only 16 of the original 32 FI items. However, the methodological sequence reported in the original study is preserved:

   (i) Random assignment yields chance level performance,

   (ii) Extracting latent frailty structure via PCA improves discrimination,

   (iii) Inclusion of age generally improved mortality prediction.

   Importantly, the cleaned dataset maintains realistic sex distributions (45.8% men, 54.2% women) and mortality proportions consistent with published estimates, confirming that preprocessing preserved demographic and structural integrity required for downstream modelling.

Chapter 5

Empirical Evaluation

You can also you this kind of referencing if needed in the text:

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions: The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to https://tilda.tcd.ie/data/accessing-data/.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

Appendix A

Python code for exporting PMF metadata to CSV

```python
import pandas as pd
import pyreadstat
from pathlib import Path
import json

data_files = {
    "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01_TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
    "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data/0053-02_TILDA_Wave2_PMF_v2.7.sav",
    "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data/0053-04_TILDA_Wave3_PMF_v3.6.sav",
    "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05_TILDA_Wave4_PMF_v4.4.sav",
    "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06_TILDA_Wave5_PMF_v5.5.sav",
    "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data/0053-03_TILDA_Harmonized_v1.2.dta",
}


def load_df_with_var_of_interest(
    var_dir="var_of_interest", data_files=data_files, verbose=False
```

```python
):
    dfs = {}
    var_dir = Path(var_dir)

    for json_name, data_path in data_files.items():
        file = var_dir / json_name
        with open(file, "r") as f:
            vars_list = json.load(f).get("variables_of_interest")
            if verbose:
                print(f"Variables of interest from {file.name}: {vars_list[:10]} ...")

        if data_path.endswith(".sav"):
            df_full, meta = pyreadstat.read_sav(
                data_path, encoding="latin1", apply_value_formats=True
            )
            df, _ = pyreadstat.read_sav(
                data_path,
                usecols=vars_list,
                encoding="latin1",
                apply_value_formats=True,
            )
        else:
            df_full = pd.read_stata(data_path, convert_categoricals=True)
            df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]

        if verbose:
            print(
                f"Data shape (Original): {df_full.shape[0]} rows x {df_full.shape[1]} columns"
            )
            print(f"Data shape (Cleaned): {df.shape[0]} rows x {df.shape[1]} columns")
            print("-----------------------------------------------")

        dfs[file.stem] = df

```

```python
    return dfs


if __name__ == "__main__":
    for json_name, path_str in data_files.items():
        path = Path(path_str)
        wave_name = json_name.replace(".json", "")
        if path_str.endswith(".sav"):
            # SPSS
            df, meta = pyreadstat.read_sav(
                path, encoding="latin1", apply_value_formats=True
            )

        else:  # Stata .dta file
            df, meta = pyreadstat.read_dta(path, encoding="latin1")

        pd.DataFrame(
            {"variable": meta.column_names, "label": meta.column_labels}
        ).to_csv(Path("output") / f"{wave_name}_variable_descriptions.csv", index=False)

        print(f"--------------- wave {path} ---------------")
        print(df.shape)
        print(df.columns[:20])
        print(
            "Variable/Feature descriptions, csv file saved successfully -> dir=output/"
        )
        df.to_csv((Path("output") / f"{wave_name}.csv"), index=False)

    dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Scenario: Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	**809**	**482**	**639**	**592**	**577**	**453**

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Appendix C

First Stage: Scenario Variable Mapping

```python
# Sentinel values (numeric and string)
# -1 and -1.0 is sentinel but will be used in the clean pipeline
standardised_sentinels = {'-1', '-1.0', -1, -1.0} #  set -> duplicates removed -> counts correct -> single value detection is zero
numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
                     95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}" for x in numeric_sentinels] + \
    ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "refuse", 'none', 'no response', 'nan', 'not answered', 'no answer',
     'not applicable','no consent','dont know',"don't know","didn't answer",'prefer not to say', 'task not done', 'insufficient sample', 'unable to record',
     '__', '*']
blanks = ['', ' ', '\t', '\n']

SCENARIOS = [
    "Empty / blank values",
    "Numeric with sentinel",
    "Clean numeric",
    "Censored / bracketed numeric",
    "Binary categorical",
```

```
    "Multi␣class␣categorical␣/␣mixed",
    "Single␣value␣categorical",
    "Special␣symbol␣/␣mixed",
    "Unknown␣/␣complex␣mixed"
]


# flatten all dtypes across the waves to float and category
def flatten_column_types(df):
    df_flat = df.copy()

    for col in df_flat.columns:
        if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric columns
            df_flat[col] = df_flat[col].astype(float)  # Convert to float
        else:
            df_flat[col] = (  # Convert to string, strip whitespace, lowercase, then category
                df_flat[col]
                .astype(str)
                .str.strip()
                .str.lower()
                .astype('category')
            )

    return df_flat

def detect_column_scenario_mapping(df, sample_size=10):
    summary = []

    # Regex patterns
    special_symbols_pattern = r'[^A-Za-z0-9\s\.,\-+&/()]' # special symbols excluding common numeric formatting
    bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[<>+-]"  # patterns like "10-20", "30/40", ">=50"
    sentinel_set = set(x.lower() for x in string_sentinels) # Lowercase string sentinels for comparison

    for col in df.columns:
```

```
        col_data = df[col].copy()
        dtype = str(col_data.dtype) # get data type as string

        # Convert to string for sentinel / blank detection
        str_vals = col_data.astype(str).str.lower().str.strip() #
            lowercase and strip whitespace
        clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(
            str_vals.isin(blanks))] # remove sentinels and blanks

        # Remove sentinels and blanks for scenario detection only
        vals_no_sentinel = clean_vals[~clean_vals.astype(str).str
            .lower().isin(sentinel_set)
                                      & ~clean_vals.astype(str).
                                          str.lower().isin(blanks)
                                          ].unique()
        num_unique_no_sentinel = len(vals_no_sentinel)

        # Check if any sentinel values are present
        sentinel_present = str_vals.isin(sentinel_set).any()

        # Detection flags
        empty_detected = str_vals.isin(blanks).any() or len(
            col_data) == 0

        # Numeric detection: coerce to numeric, NaNs for non-
            numeric
        numeric_vals = pd.to_numeric(clean_vals, errors='coerce')
             # coerce non-numeric to NaN
        numeric_like = numeric_vals.notna().all() and len(
            clean_vals) > 0 # all values are numeric-like after
            cleaning

        sentinel_detected = str_vals[~str_vals.isin(
            standardised_sentinels)].isin(sentinel_set).any() # or
             col_data.eq(-1.0).any() # check for sentinel presence
        censored_detected = clean_vals.astype(str).str.contains(
            bracketed_numeric_pattern).any() # check for bracketed
             numeric patterns

        # Binary categorical detection (exactly 2 unique values
            after cleaning)
```

```python
        binary_categorical_detected = False
        if dtype == 'category':
            if num_unique_no_sentinel == 2:
                binary_categorical_detected = True
            elif num_unique_no_sentinel == 1 and sentinel_present:
                binary_categorical_detected = True

        # Multi-class categorical (3+ unique values, object/category)
        multi_class_categorical_detected = num_unique_no_sentinel > 2 and dtype == 'category'

        # Single value categorical
        single_value_detected = num_unique_no_sentinel == 1 and not binary_categorical_detected

        # Special symbols detection (ignore numeric formatting)
        symbol_detected = clean_vals.astype(str).str.contains(special_symbols_pattern).any()

        # Mixed types detection
        mixed_detected = len(set(type(v) for v in col_data)) > 1  # more than one data type present

      # Determine scenario in order of pipeline
        if empty_detected:
            scenario = SCENARIOS[0]
            handling = "Replace numeric/categorical empty/blank with -1.0"
        elif numeric_like and sentinel_detected:
            scenario = SCENARIOS[1]
            handling = "Replace numeric sentinel values with -1.0"
        elif numeric_like and not sentinel_detected:
            scenario = SCENARIOS[2]
            handling = "Keep as numeric"
        elif censored_detected:
            scenario = SCENARIOS[3]
            handling = "Convert numeric ranges to midpoints or keep as categorical"
```

```
        elif binary_categorical_detected:
            scenario = SCENARIOS[4]
            handling = "Encode as 0/1 or keep as category"
        elif multi_class_categorical_detected:
            scenario = SCENARIOS[5]
            handling = "One-hot encode, label encode, or keep as category"
        elif single_value_detected:
            scenario = SCENARIOS[6]
            handling = "Keep as category / constant column"
        elif symbol_detected:
            scenario = SCENARIOS[7]
            handling = "Inspect and preprocess based on context"
        else:
            scenario = SCENARIOS[8]
            handling = "Inspect manually / complex mixed types"

        sample_values = list(col_data.unique()[:sample_size]) # sample unique values

        summary.append({
            'Variable': col,
            'DataType': dtype,
            'UniqueValues': num_unique_no_sentinel,
            'Scenario': scenario,
            'SuggestedHandling': handling,
            'SampleValues': sample_values
        })

    return pd.DataFrame(summary)
```

Appendix D

Step 6: Scenario ‘Multi class categorical’ Cleaning Function

```python
def categorical_transform_S6(df): # Step6; Scenario -> Multi class categorical
    df_cleaned = df.copy()
    sent_set = set([str(x).lower() for x in string_sentinels]) | set([str(x).lower() for x in standardised_sentinels])
    numeric_sentinel_set = set(float(x) for x in numeric_sentinels)

    # Helper normaliser
    def _norm(tok):
        if tok is None:
            return None
        s = str(tok).strip().lower()
        if s in blanks:
            return ''
        return s

    # Words -> number map
    word_num_map = {
        'zero':0.0,'one':1.0,'two':2.0,'three':3.0,'four':4.0,'five':5.0,'six':6.0,'seven':7.0,'eight':8.0,'nine':9.0,
        'ten':10.0,'eleven':11.0,'twelve':12.0,'thirteen':13.0,'fourteen':14.0,'fifteen':15.0,'twenty':20.0,
```

```
        'thirty':30.0,'forty':40.0,'fifty':50.0,'sixty':60.0,'seventy':70.0,'eighty':80.0,'ninety':90.0,
        'a few':2.0,'a couple':2.0,'few':2.0,'once':1.0,'twice':2.0,'several':3.0,'couple':2.0
    }
    def words_to_num(token):
        if token is None:
            return None
        token = token.strip().lower()
        if token in word_num_map:
            return word_num_map[token]
        try:
            return float(token)
        except Exception:
            return None

    # Regexes for parsing
    RE_TIME = re.compile(r'^\s*(\d{1,2}):(\d{2})\s*$')
    RE_PERCENT = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*%\s*$')
    RE_DIRECT_NUM = re.compile(r'^\s*-?\d+(?:\.\d+)?\s*$')
    RE_RANGE = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*[---/]\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_RANGE_TO = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*(?:to|TO)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_PLUS = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\s*\+\s*$')
    RE_GT = re.compile(r'^\s*(?:>=|>)\s*(-?\d+(?:\.\d+)?)\s*$')
    RE_LT_ANCHOR = re.compile(r'^(?:<|<=)\s*-?\d+(?:\.\d+)?')
    RE_NUM_FIND = re.compile(r'-?\d+(?:\.\d+)?')
    RE_WORD_TOK = re.compile(r"[a-z']+(?:\s[a-z']+)?")
    # RE_SURVEY_TIME = re.compile(r'^\s*(?:[a-z]+\s+)?(\d+(?:\.\d+)?)\s*(?:month|week|year)s?.*$', re.IGNORECASE)

    def norm_return(v):
        try:
            vf = float(v)
        except Exception:
            return None
        if vf in numeric_sentinel_set:
            return 'SENTINEL'
        return vf

```

```
    def try_parse_token(x):
        """Return float, 'SENTINEL', or None for unknown token x"""
        if x is None:
            return None

        # preserve numeric pre transformed values
        if isinstance(x, (int, float)):
            if float(x) in numeric_sentinel_set:
                return 'SENTINEL'
            return float(x)

        s = _norm(x)
        # Extending sentinel detection for categorical parsing scenario
        if s == '':
            return 'SENTINEL'
        if 'not applicable' in s or 'does not have' in s or "don't receive" in s or "don\x92t receive" in s or 'unknown' in s:
            return 'SENTINEL'
        if s in sent_set or re.fullmatch(r'[^0-9a-z]+', s):
            return 'SENTINEL'
        if s in ('other', 'other specify') or s.startswith('other (') or s.endswith('(specify)'):
            return 'SENTINEL'

        s0 = s.replace(',', '').replace('$', '').replace(EUR, '').replace('#', '')

        s0 = re.sub(r'(\d+)-', r'\1 -', s0)
        s0 = re.sub(r'-(\d+)', r'- \1', s0)

        RE_PREFIX_NUM = re.compile(r'^\s*(-?\d+(?:\.\d+)?)\.\s*.+$')

        m = RE_PREFIX_NUM.match(s0)
        if m:
            # If the format is 'NUM. anything', return the number as the category value
            return norm_return(float(m.group(1)))
```

```
        # time
        m = RE_TIME.match(s0)
        if m:
            h, mi = float(m.group(1)), float(m.group(2))
            return norm_return(h * 60.0 + mi)

        # percent
        m = RE_PERCENT.match(s0)
        if m:
            return norm_return(float(m.group(1)))

        # direct numeric
        if RE_DIRECT_NUM.match(s0):
            return norm_return(s0)

        # ranges
        m = RE_RANGE.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass
        m = RE_RANGE_TO.match(s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # plus / greater than
        m = RE_PLUS.match(s0) or RE_GT.match(s0)
        if m:
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # less than
        if RE_LT_ANCHOR.search(s0) or 'less than' in s0:
```

```
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                return norm_return(nums[0])

        # or less / more
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*less', s0)
        if m:
            return norm_return(m.group(1))
        m = re.search(r'(-?\d+(?:\.\d+)?)\s*or\s*more', s0)
        if m:
            return norm_return(m.group(1))

        # "more than X but less than Y", "over X but less than Y"
        m = re.search(r'(?:more than|over)\s*e?(-?\d+(?:\.\d+)?)\s*(?:but\s*)?(?:less than|under)\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # "between X and Y" / "between X - Y"
        m = re.search(r'between\s*e?(-?\d+(?:\.\d+)?)\s*(?:and|[---/])\s*e?(-?\d+(?:\.\d+)?)', s0)
        if m:
            try:
                a, b = float(m.group(1)), float(m.group(2))
                return norm_return((a + b) / 2.0)
            except Exception:
                pass

        # survey time based
        time_map_weeks = {
            'one week': 1.0,
            'two weeks': 2.0,
            'a month (4 weeks)': 4.0,
            'a month/4 weeks': 4.0,
            'three months (13 weeks)': 13.0,
            'three months/13 weeks': 13.0,
```

```
            'six months (26 weeks)': 26.0,
            'six months/26 weeks': 26.0,
            'one year (12 months/52 weeks)': 52.0,
            'one year/12 months/52 weeks': 52.0
        }
        if s0 in time_map_weeks:
            return norm_return(time_map_weeks[s0])

        # If a category is '3 months or more', try to extract the number and use upper bound
        if 'months or more' in s0 and RE_NUM_FIND.search(s0):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                months = float(nums[0])
                return norm_return(months * 4.33) # Convert months to weeks (4.33 weeks per month)

        # Frequency calendar based
        if any(tok in s0 for tok in ('day','week','month','year','daily','never','almost every day','once','twice','thrice','three','several','couple')):
            nums = RE_NUM_FIND.findall(s0)
            if nums:
                val = (float(nums[0]) + float(nums[1])) / 2.0 if len(nums) >= 2 else float(nums[0])
            else:
                if 'once or twice' in s0:
                    val = 1.5
                else:
                    val = None
                    for w in RE_WORD_TOK.findall(s0):
                        wn = words_to_num(w)
                        if wn is not None:
                            val = wn
                            break
            if val is not None:
                if 'day' in s0:
                    if 'week' in s0:
                        return norm_return(val * 4.33)
                    if 'month' in s0:
                        return norm_return(val)
```

```
                    if 'daily' in s0 or 'every day' in s0 or 'almost every day' in s0:
                        return norm_return(val * 30.0)
                    return norm_return(val * 4.33)
                if 'week' in s0:
                    return norm_return(val * 4.33)
                if 'month' in s0:
                    return norm_return(val)
                if 'year' in s0:
                    return norm_return(val / 12.0)
                return norm_return(val)


        likert_map = {
            # Likert / ordinals (expanded with quintile/lowest/highest detection)
            'strongly disagree': 0.0, 'disagree': 1.0, 'neither agree nor disagree': 2.0, 'neither': 2.0,
            'agree': 3.0, 'strongly agree': 4.0,
            'very poor': 0.0, 'poor': 1.0, 'fair': 2.0, 'good': 3.0, 'very good': 4.0, 'excellent': 5.0,
            'none': 0.0, 'not at all': 0.0, 'hardly ever': 0.0,
            'a little': 1.0, 'some': 2.0, 'a lot': 3.0, 'alot': 3.0, 'often': 3.0,
            'yes': 1.0, 'no': 0.0, 'sometimes': 2.0, 'rarely': 1.0, 'most or all of the time': 4.0,
            'pass': 2.0, 'fail': 1.0, '100': 100.0,
            # Problem severity mappings
            'no problem': 1.0,'minor problem': 2.0,'moderate problem': 3.0,'major problem': 4.0,
            # Concern level mappings
            'not at all concerned': 1.0,'somewhat concerned': 2.0,'fairly concerned': 3.0,'very concerned': 4.0,
            # Survey frequency mappings
            'daily / almost daily': 7.0, 'once a week or more': 6.0,'twice a month or more': 5.0,'about once a month': 4.0,
            'every few months': 3.0, 'about once or twice a year': 2.0,'less than once a year': 1.0,'never': 0.0,
            'daily': 7.0,'every day': 7.0,'almost every day': 7.0,'most or all of the time': 7.0,
```

```
            'every␣week': 6.0,'once␣a␣week': 6.0,'several␣times␣per␣week': 6.0,
            'twice␣a␣month': 5.0,'two␣a␣month': 5.0,'several␣times␣per␣month': 5.0,'once␣or␣twice␣a␣month': 5.0,
            'once␣a␣month': 4.0, 'once␣or␣twice␣a␣year': 2.0,'less␣than␣once␣a␣month': 1.0,
        }

        if s0 in likert_map:
            return float(likert_map[s0])

        # Quintile / ordinal like tokens: 'lowest', '2nd quintile', 'highest'
        if 'quintile' in s0 or 'lowest' in s0 or 'highest' in s0 or 'middle' in s0 or 'median' in s0:
                if 'lowest' in s0:
                    return norm_return(1.0)
                if 'highest' in s0:
                    return norm_return(5.0)
                if 'middle' in s0 or 'median' in s0:
                    return norm_return(3.0)
                mq = re.search(r'(\d+)(?:st|nd|rd|th)?\s+quintile', s0)
                if mq:
                    return norm_return(float(mq.group(1)))
                nums = RE_NUM_FIND.findall(s0)
                if nums:
                    return norm_return(nums[0])

        if 'adjacent' in s0: # Adjacent boxes: keep semantic average if present
            nums = RE_NUM_FIND.findall(s0)
            if len(nums) >= 2:
                return norm_return((float(nums[0]) + float(nums[1])) / 2.0)
            elif nums:
                return norm_return(nums[0])
            return None

        # Fallback: first number
```

```python
        nums = RE_NUM_FIND.findall(s0)
        if nums:
            return norm_return(nums[0])

        return None

    def category_sort_key(u):
        us = _norm(u) or ''
        parsed = try_parse_token(us)
        if isinstance(parsed, (int, float)):
            return (0, float(parsed), us)
        if us == 'no' or us.startswith('no ') or us.startswith('not '):
            return (1, 0.0, us)
        if 'yes' in us:
            if 'both' in us or 'two' in us:
                return (1, 2.0, us)
            if 'one' in us:
                return (1, 1.0, us)
            return (1, 1.0, us)
        amount_keywords = ['none', 'not at all', 'hardly ever', 'a little', 'some', 'a lot', 'alot', 'often', 'sometimes', 'rarely']
        for i, key in enumerate(amount_keywords):
            if key in us:
                return (2, float(i), us)
        if 'adjacent boxes' in us or 'adjacent numbers' in us:
            nums = RE_NUM_FIND.findall(us)
            if len(nums) >= 2:
                a, b = float(nums[0]), float(nums[1])
                return (10, (a + b) / 2.0, us)
            elif nums:
                return (10, float(nums[0]), us)
            return (10, 0, us)
        return (99, 0, us)

    # Operate only on categorical columns
    cat_cols = df_cleaned.select_dtypes(include='category').columns.tolist()
    for col in cat_cols:
```

```python
        uniques = list(dict.fromkeys(df_cleaned[col].tolist())) # Preserve original tokens with types (do NOT cast to str here)
        normalised_pairs = [(u, _norm(u)) for u in uniques] # Build normalised pairs keeping original token too
        # Keep tokens that are non sentinel and parseable (strings or numbers)
        non_sentinel = [u for u, nu in normalised_pairs if (nu and nu not in sent_set and nu not in blanks and try_parse_token(nu) != 'SENTINEL')]

        if not non_sentinel:
            continue

        # Computing parse results keyed by normalized string (but keep list of originals too)
        parse_results = { _norm(u): try_parse_token(_norm(u)) for u in non_sentinel }

        numeric_count = sum(1 for v in parse_results.values() if isinstance(v, (int, float)))
        sentinel_parsed_count = sum(1 for v in parse_results.values() if v == 'SENTINEL')
        all_parse_numeric = (numeric_count + sentinel_parsed_count == len(parse_results))

        if all_parse_numeric:
            # Map token string -> numeric (SENTINEL -> -1.0)
            code_map = { k: (float(v) if isinstance(v, (int, float)) else -1.0) for k, v in parse_results.items() }

            def map_to_numeric(x):
                # Preserve the already numeric cells (except numeric sentinels)
                if isinstance(x, (int, float)) and not pd.isna(x):
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
```

```
                    return -1.0
                xs = _norm(x)
                if not xs or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0
                return float(code_map.get(xs, -1.0))

            df_cleaned[col] = df_cleaned[col].apply(map_to_numeric).astype(float)

        else:
            first_seen_norm = []
            seen = set()
            for u in non_sentinel:
                nu = _norm(u)
                if nu not in seen:
                    first_seen_norm.append(nu)
                    seen.add(nu)

            parse_results = {nu: try_parse_token(nu) for nu in first_seen_norm}
            max_reserved_val = -1.0
            for nu, val in parse_results.items():
                if isinstance(val, (int, float)):
                    max_reserved_val = max(max_reserved_val, val) # Assuming any parsed float is a reserved ID if it came from the map
            start_i = int(max_reserved_val) + 1
            if start_i == 0: # Ensure we never start at 0 if 0 is a reserved category i.e for ('no')
                start_i = 1
            string_tokens = [nu for nu in first_seen_norm if not isinstance(parse_results.get(nu), (int, float))]
            sorted_strings = sorted(string_tokens, key=category_sort_key)

            code_map_strings = { norm_tok: i for i, norm_tok in enumerate(sorted_strings, start=start_i) }

            def map_mixed(x):
```

```python
                if isinstance(x, (int, float)) and not pd.isna(x):# Preserve real numeric cell values
                    if float(x) in numeric_sentinel_set:
                        return -1.0
                    return float(x)
                if pd.isna(x):
                    return -1.0
                xs = _norm(x)
                if xs == '' or xs in sent_set or xs in blanks or re.fullmatch(r'[^0-9a-z]+', xs):
                    return -1.0

                parsed_val = try_parse_token(xs)
                if parsed_val == 'SENTINEL':
                    return -1.0
                if isinstance(parsed_val, (int, float)):# Captures 'no' (0.0), 'yes' (1.0), and other hardcoded likert values.
                    return float(parsed_val)
                if xs in code_map_strings:
                    return float(code_map_strings[xs])
                return -1.0

            df_cleaned[col] = df_cleaned[col].apply(map_mixed).astype(float)

    # Final sweep on numeric columns
    for c in df_cleaned.select_dtypes(include='float').columns:
        df_cleaned[c] = df_cleaned[c].fillna(-1.0)
        df_cleaned[c] = df_cleaned[c].replace(list(numeric_sentinel_set), -1.0)

    return df_cleaned
```

Appendix E

Second Stage: Deterministic *six step* Cleaning Pipeline

```
# Cleaning pipeline to all waves
dfs_cleaned = {}

# flattened dtype df
dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
    items()} # flatten dtypes

# Original df
display(scenario_count_table(dfs_flat).style.set_caption("
    Original Table - pre cleaning")) # display original scenario
    count table

# Step 1: Clean empty/missing values
dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
    dfs_flat.items()} # apply step 1 cleaning
display(scenario_count_table(dfs_step1).style.hide(axis="index").
    set_caption("Step 1: Empty / Blank - post cleaning"))

# Step 2: Clean sentinel values
dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
    items()} # apply step 2 cleaning
display(scenario_count_table(dfs_step2).style.hide(axis="index").
    set_caption("Step 2: Sentinel - post cleaning"))

```

```
# Step 3: Numerical Transformation
dfs_step3 = {wave: numerical_transform(df) for wave, df in
    dfs_step2.items()} # apply step 3 cleaning
display(scenario_count_table(dfs_step3).style.hide(axis="index").
    set_caption("Step 3: Numeric Transformation - post cleaning"))

# Step 4: Categorical Transformation for censored scenario
dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in
    dfs_step3.items()} # apply step 4 cleaning
display(scenario_count_table(dfs_step4).style.hide(axis="index").
    set_caption("Step 4: Category Transformation S4 - post 
    cleaning"))

# Step 5: Categorical Transformation for binary values
dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in
    dfs_step4.items()} # apply step 5 cleaning
display(scenario_count_table(dfs_step5).style.hide(axis="index").
    set_caption("Step 5: Category Transformation S5 - post 
    cleaning"))

# Step 6: Categorical Transformation for multi class/mixed values
dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in
    dfs_step5.items()} # apply step 6 cleaning
display(scenario_count_table(dfs_step6).style.hide(axis="index").
    set_caption("Step 6: Category Transformation S6 - post 
    cleaning"))

dfs_cleaned = dfs_step6
```

Appendix F

MD Dictionary: Scenario Grouping and Transformation Function

```
def show_grouped_vars_for_scenarios(dfs_original, dfs_transformed, scenarios_to_show): # Function to show grouped variables per scenario across waves -> markdown format
    """
    Used to inspect the scenario variables/features that need handling for transformation/cleaning.
    A pre and post cleaning helper function to identify variables that need specific handling and to
    Inspect actual changes for a scenario from previous cleaning step to current step.
    """

    print("# Category Transformation Mapping")
    # print(f"\n## Scenarios: {scenarios_to_show}") # Markdown mapping dictionary header
    for wave in dfs_original.keys():
        print(f"\n### ==================== {wave.upper()} ====================")

        mapping_df = detect_column_scenario_mapping(dfs_transformed[wave]) # Get scenario mapping from transformed/cleaned to see overall changes for all variables
```

```
    # mapping_df = detect_column_scenario_mapping(dfs_original[wave]) # Get scenario mapping from original to see untransformed values

    for scenario in scenarios_to_show:
        vars_in_scenario = mapping_df[mapping_df['Scenario'] == scenario]['Variable'].tolist() # Variables in scenario
        print(f"\n--- Scenario {scenario}: {len(vars_in_scenario)} ---") # Scenario header for pipeline git commit
        if not vars_in_scenario:
            print("\nNo variables found.")
            continue

        # Dictionary: frozen set(values) -> list of variables
        groups = {}

        for var in vars_in_scenario:
            unique_vals = list(dict.fromkeys(dfs_original[wave][var].dropna()))  # Preserve first seen order from ORIGINAL
            key = tuple(unique_vals)  # Use ordered tuple as key

            if key not in groups: # Initialise list if key not present
                groups[key] = []
            groups[key].append(var) # Append variable to the group

        # Print grouped variables in markdown format friendly for Notion -> mapping dictionary
        for value_set, var_list in groups.items():
            print("\n*Variables:*","`" + ", ".join(var_list) + "`  ") # \n
            print(f"*Pre transformed values:* {list(value_set)}  ")

            # Pull transformed values from dfs_cleaned / specific step using preserved order
```

```python
                try:
                    df_before = dfs_original[wave][var_list[0]]
                    df_after = dfs_transformed[wave][var_list[0]]

                    # Build transformed list in same order as original values
                    transformed_vals = []
                    for orig_val in value_set: # Iterate through original values in order
                        mask = df_before == orig_val
                        if mask.any():
                            trans_vals = df_after[mask].unique().tolist() # Get unique transformed values
                            if trans_vals:
                                transformed_vals.append(trans_vals[0]) # Append first transformed value
                except Exception:
                    transformed_vals = []
                print(f"*Transformed numeric:* `{transformed_vals}`")


scenarios_to_show = SCENARIOS
# scenarios_to_show = ["Clean numeric"]
# scenarios_to_show = ["Censored / bracketed numeric"]
# scenarios_to_show = ["Binary categorical"]
# scenarios_to_show = ["Multi class categorical / mixed"]

show_grouped_vars_for_scenarios(dfs, dfs_cleaned, scenarios_to_show) # prev step == comparison to current step else current step transformed/cleaned for ready .md dictionary
```

Appendix G

MD Dictionary: Scenario Grouping and Transformation Audit Outputs

- `sex`: Binary biological sex indicator (e.g., `Male/Female`, `Female/Male`, prefixed encodings).
- `ph001`: Self-rated general health status measured on an ordinal scale. Survey question: *"Now I would like to ask you some questions about your health. Would you say your health is. . . "*. Responses range from *Poor* to *Excellent*, with additional 'no response' categories.
- `ph009`: Comparative self-rated health relative to people of the same age. Survey question: *"In general, compared to other people your age, would you say your health is. . . "*.
- `bh006`: Frequency based lifestyle variable combining numeric values, bounded ranges (e.g., `20-39`), and textual descriptors (e.g., `Less than 10`, `30+`). Survey question: *"How many cigarettes do/did you smoke on average per day?"*.

```
%-----------------------------------------------------------------------%
%------------------------MARKDOWN_MAPPING_DICTIONARY----------------------%
%-----------------------------------------------------------------------%
# Category Transformation Mapping
```

G. MD DICTIONARY: SCENARIO GROUPING AND TRANSFORMATION AUDIT OUTPUTS

=================== WAVE1 ===================

--- Scenario Empty / blank values: 0 ---

No variables found.

--- Scenario Numeric with sentinel: 0 ---

No variables found.

--- Scenario Clean numeric: 809 ---

Variables: ‘sex, gd002‘
Pre transformed values: [’Male’, ’Female’]
Transformed numeric: ‘[1.0, 0.0]‘

Variables: ‘ph001‘
Pre transformed values: [’Fair’, ’Good’, ’Very good’, ’Excellent’, ’Poor’, "Don’t k
Transformed numeric: ‘[2.0, 3.0, 4.0, 5.0, 1.0, -1.0]‘

Variables: ‘bh006‘
Pre transformed values: [’20-39’, 10.0, -1.0, 15.0, 5.0, ’60+’, "Don’t Know", 2.0,
Transformed numeric: ‘[29.5, 10.0, -1.0, 15.0, 5.0, 60.0, -1.0, 2.0, 8.0, 3.0, 49.5

--- Scenario Censored / bracketed numeric: 0 ---

No variables found.

--- Scenario Binary categorical: 0 ---

No variables found.

--- Scenario Multi class categorical / mixed: 0 ---

```
No variables found.

--- Scenario Single value categorical: 0 ---

No variables found.

--- Scenario Special symbol / mixed: 0 ---

No variables found.

--- Scenario Unknown / complex mixed: 0 ---

No variables found.

### =================== WAVE2 ====================

...

--- Scenario Clean numeric: 482 ---

*Variables:* ‘gd002, sex‘
*Pre transformed values:* [’Female’, ’Male’]
*Transformed numeric:* ‘[0.0, 1.0]‘

*Variables:* ‘bh006‘
*Pre transformed values:* [-1.0, ’10 - 19’, ’20 - 29’, ’Less than 10’, ’30+’,
*Transformed numeric:* ‘[-1.0, 14.5, 24.5, 10.0, 30.0, -1.0]‘

*Variables:* ‘ph001‘
*Pre transformed values:* [’Very good’, ’Good’, ’Fair’, ’Excellent’, ’Poor’]
*Transformed numeric:* ‘[4.0, 3.0, 2.0, 5.0, 1.0]‘

...
```

G. MD DICTIONARY: SCENARIO GROUPING AND TRANSFORMATION AUDIT OUTPUTS

```
### =================== WAVE3 ===================

...

--- Scenario Clean numeric: 639 ---

*Variables:* ‘sex, gd002‘
*Pre transformed values:* [’Female’, ’Male’]
*Transformed numeric:* ‘[0.0, 1.0]‘

*Variables:* ‘ph001, ph009‘
*Pre transformed values:* [’Very Good’, ’Good’, ’Excellent’, ’Fair’, ’Poor’, ’DK’]
*Transformed numeric:* ‘[4.0, 3.0, 5.0, 2.0, 1.0, -1.0]‘

*Variables:* ‘bh006‘
*Pre transformed values:* [-1.0, ’10 - 19’, ’Less than 10’, ’20 - 29’, ’30+’, -98.0,
*Transformed numeric:* ‘[-1.0, 14.5, 10.0, 24.5, 30.0, -1.0, -1.0]‘

...

### =================== WAVE4 ===================

...

--- Scenario Clean numeric: 592 ---

*Variables:* ‘bh006‘
*Pre transformed values:* [-1.0, ’Less than 10’, ’20 - 29’, ’10 - 19’, ’30+’, "Don’t
*Transformed numeric:* ‘[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]‘

*Variables:* ‘gd002, sex‘
*Pre transformed values:* [’Female’, ’Male’]
*Transformed numeric:* ‘[0.0, 1.0]‘
```

```
*Variables:* ‘ph001‘
*Pre transformed values:* [’Good’, ’Fair’, ’Poor’, ’Excellent’, ’Very Good’, ’
*Transformed numeric:* ‘[3.0, 2.0, 1.0, 5.0, 4.0, -1.0, -1.0]‘

...

### ==================== WAVE5 ====================

...

--- Scenario Clean numeric: 577 ---

*Variables:* ‘bh006‘
*Pre transformed values:* [-1.0, ’Less than 10’, ’20 - 29’, ’10 - 19’, ’30+’,
*Transformed numeric:* ‘[-1.0, 10.0, 24.5, 14.5, 30.0, -1.0]‘

*Variables:* ‘gd002, sex‘
*Pre transformed values:* [’Female’, ’Male’]
*Transformed numeric:* ‘[0.0, 1.0]‘

*Variables:* ‘ph001‘
*Pre transformed values:* [’Very Good’, ’Fair’, ’Excellent’, ’Good’, ’Poor’]
*Transformed numeric:* ‘[4.0, 2.0, 5.0, 3.0, 1.0]‘

...
```

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